

mm-Wave Phase Control

Prior-art baseline comparison and gap analysis

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2026-05-04

This work: 35 GHz half-rate PI $\rightarrow \times 2$
doubler $\rightarrow$ 70 GHz

Phase-error doubling rule:

$$\epsilon_{\phi,70} = 2 \epsilon_{\phi,35}$$

Phase noise +6 dB ($20 \log_{10} 2$); absolute jitter preserved.

Baseline families — 8 works

- Active vector-modulator PS (15–90 GHz)
- Passive RTPS (28 GHz)
- Passive switched APN (14–50 GHz)
- Passive PI in T/R channels (W-band)

Quantitative Metrics

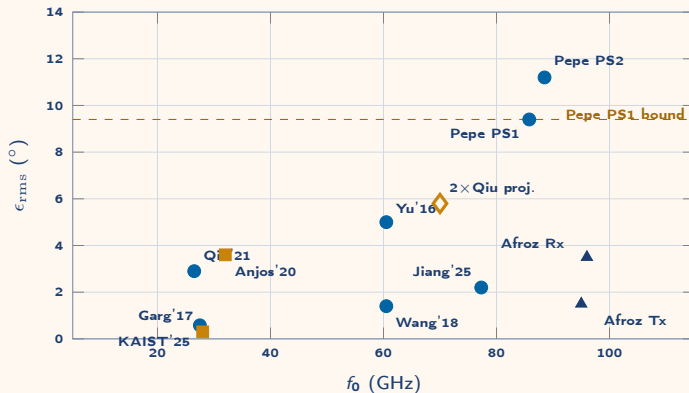
Work	f_0 (GHz)	FBW (%)	Bits	LSB ($^\circ$)	ϵ_{rms} ($^\circ$)	ϵ/LSB
Qiu et al. 2021	26.5	86.8	6	5.63	2.23–3.5	0.40–0.62
Pepe/Zito PS1	85.8	16.3	4	22.5	< 11.9	< 0.53
Pepe/Zito PS2	88.5	18.8	4	22.5	< 11.9	< 0.53
Yu et al. 2016	60.5	11.6	4	22.5	2.3–7.6 ^a	0.10–0.34
Afroz et al. Rx	96.0	8.3	4	22.5	< 6	< 0.27
Afroz et al. Tx	95.0	21.1	4	22.5	< 1.5 / < 8 ^b	< 0.07 / < 0.36
Garg/Natarajan	28.0	14.3	5	11.25	< 1	< 0.089
Anjos et al. 2020	32.0	112.5	2	45	3.6 / 5.9 ^c	0.08 / 0.13

^aAfter calibration. ^b $< 8^\circ$ near Tx saturation. ^cAt 28 GHz / 39 GHz.

Work	Architecture	Gain / IL (dB)	DC Power	NF (dB)	$P_{1\text{dB}}$ (dBm)
Qiu'21	Active VM	active gain	19.2 mW	—	—
Pepe PS1	Active VM	+2.3	21.6 mW	10.8	−7
Pepe PS2	Active VM	+0.83	21.6 mW	11.9	−6
Yu'16	Active VM	−0.4 to +2.5	19.8 mW	11 ± 1.3	−9.8
Afroz Rx	Passive PI (Rx)	+17 to +23	18 mW	5.8–6.6	−31
Afroz Tx	Passive PI (Tx)	+7.5 to +12	26–50 mW	—	—
Garg'17	Passive RTPS	−7.75	0 ^d	< 5.5	−22
Anjos'20	Passive APN	−7.4 to −10	0 (passive)	—	—

^dRTPS zero dc; full Rx front end 10 mW (LNA included).

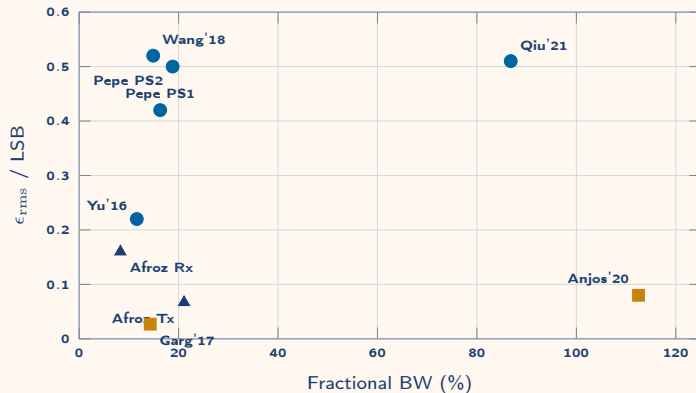
Visual Analysis



- Active VM
- ▲ Passive PI ch.
- Passive PS
- ◇ Projection ($\times 2$)

$\epsilon_{\phi,70} = 2\epsilon_{\phi,35}$, ideal doubling, from Qiu'21 mid-band.

Dashed = Pepe PS1 center-frequency error.



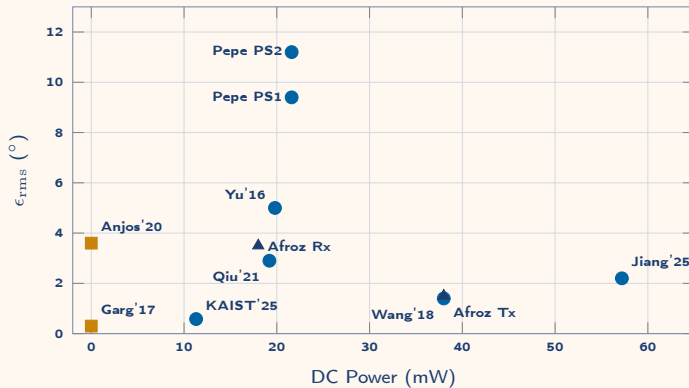
- Active VM
- ▲ Passive PI ch.
- Passive PS

Lower-left = narrow BW and high precision.

Active VMs cluster at mid-range ϵ/LSB .

Passive PSs split: high FBW (Anjos'20) vs. very low ϵ/LSB (Garg'17).

Pareto Front



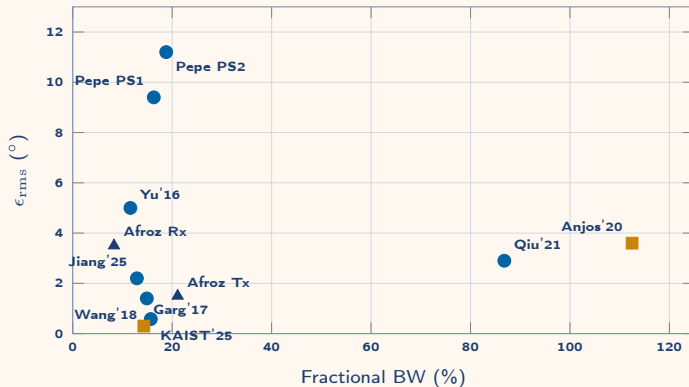
● Active VM

▲ Passive PI ch.

■ Passive PS

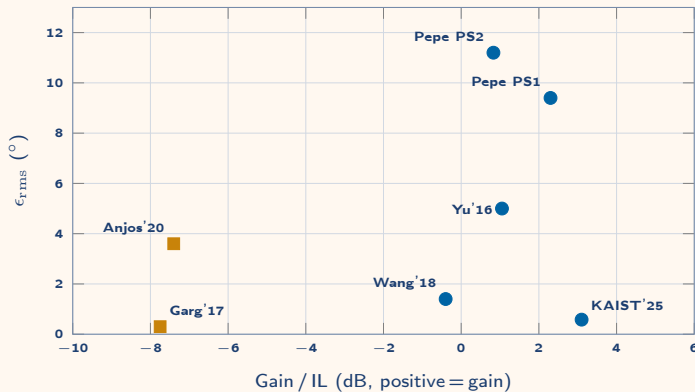
Garg'17, Anjos'20 at zero dc.

Sub-3° at >60 GHz costs > 38 mW
(Jiang'25).



- Active VM
- ▲ Passive PI ch.
- Passive PS

Jiang, Wang, KAIST cluster at
13–16% FBW, $\epsilon < 3^\circ$.



● Active VM

■ Passive PS

Passive PSs pay 7–8 dB IL.

Active VMs sit near 0 dB gain.

Afroz Rx/Tx, Qiu'21, Jiang'25

excluded (see note).

Gap Analysis

Work	f_0 (GHz)	Architecture	Key gap for 35 GHz→70 GHz PI
Qiu'21	26.5	Active VM (broadband)	Unproven above 40 GHz; NF / P1dB not reported
Pepe PS1/PS2	85–89	Active VM (W-band)	$\sim 10^\circ$ phase error; NF 11 dB; P1dB < -6 dBm
Yu'16	60.5	Active VM (60 GHz)	Needs calibration; NF ~ 11 dB; P1dB -10 dBm
Afroz Rx	96.0	Passive PI (Rx ch.)	Embedded in full Rx chain; 8 GHz phase window
Afroz Tx	95.0	Passive PI (Tx ch.)	Embedded in full Tx chain; degrades near saturation
Garg'17	28.0	Passive RTPS	~ 8 dB insertion loss; not a PI; not broadband
Anjos'20	32.0	Passive APN	Only 2-bit (45° LSB); 7–10 dB IL; poor scalability

vs. active vector-modulator baselines

- **W-band VM (Pepe):** $\epsilon_{\phi,35} < 4.5^\circ + \text{low doubler AM-PM} \Rightarrow \epsilon_{\phi,70} < 9^\circ$ (beats Pepe PS1/PS2)
- **V-band VM (Yu):** $\epsilon_{\phi,35} < 1^\circ$ after calibration (beats Yu'16 calibrated floor)

vs. passive baselines

- **RTPS / APN:** avoid 7–10 dB loss at final carrier; emphasize clock/jitter metrics
- **W-band passive PI:** present standalone PI block, not a result embedded inside full T/R channel

Questions